CONHECENDO OS COMPONENTES GRÁFICOS

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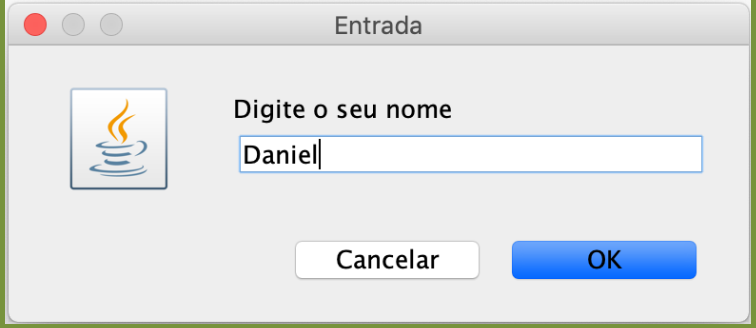
FEEVALE DIGITAL

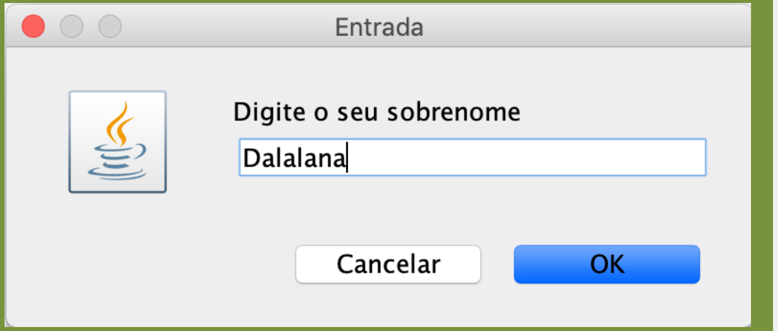
GUI (*GRAPHICAL USER INTERFACE)*

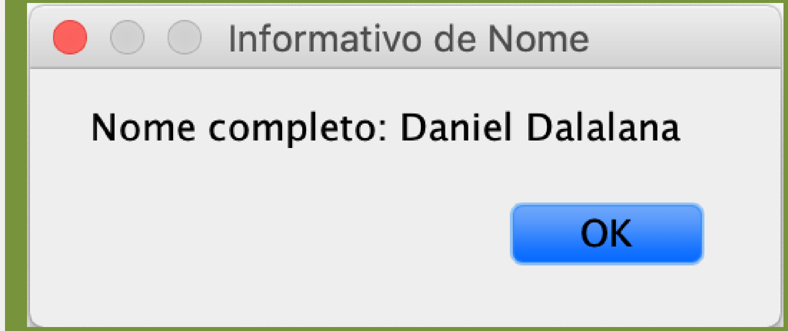
APLICAÇÃO <-> USUÁRIO

EXEMPLO 1

```
import javax.swing.JOptionPane;
public class R1_Exemplo1 {
  public static void main(String[] args) {
          String meuNome;
          String meuSobrenome;
          meuNome = JOptionPane.showInputDialog("Digite o seu nome");
          meuSobrenome = JOptionPane.showInputDialog("Digite o seu sobrenome");
          JOptionPane.showMessageDialog(null, "Nome completo: " + meuNome + " " + meuSobrenome, "Informativo de Nome", JOptionPane.PLAIN_MESSAGE);
  }
}
```







EXEMPLO 2

```
import java.awt.FlowLayout;
import javax.swing.JFrame;
import javax.swing.JLabel;

public class R1_Exemplo2 extends JFrame {

        private final JLabel minhaLabel1;
        private final JLabel minhaLabel2;

        public R1_Exemplo2(){

                setLayout(new FlowLayout());

                minhaLabel1 = new JLabel("Minha label 1");
                minhaLabel1.setToolTipText("Essa é a minha label1");
                add(minhaLabel1);

                minhaLabel2 = new JLabel("Minha label 2");
                minhaLabel2.setToolTipText("Essa é a minha label2");
                add(minhaLabel2);

        }

}
```

```
import javax.swing.JFrame;

public class R1_Exemplo2_teste {

        public static void main(String[] args) {
                R1_Exemplo2 exemplo2 = new R1_Exemplo2();

        exemplo2.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
                exemplo2.setSize(90,180);
                exemplo2.setVisible(true);
        }

}
```

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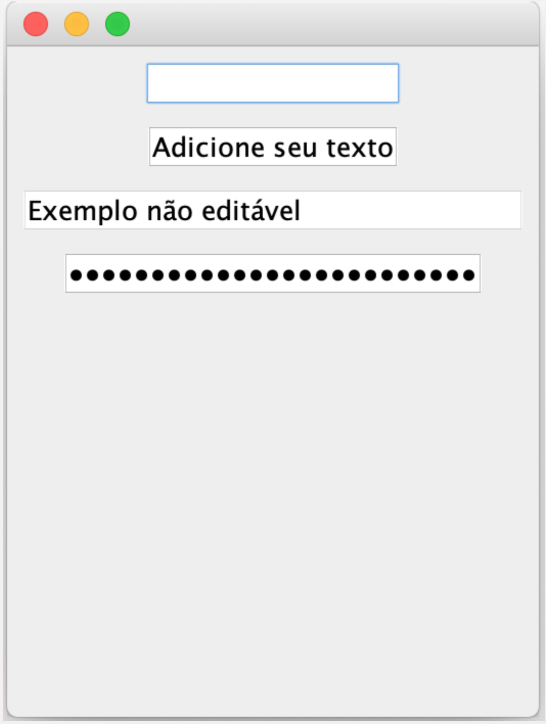
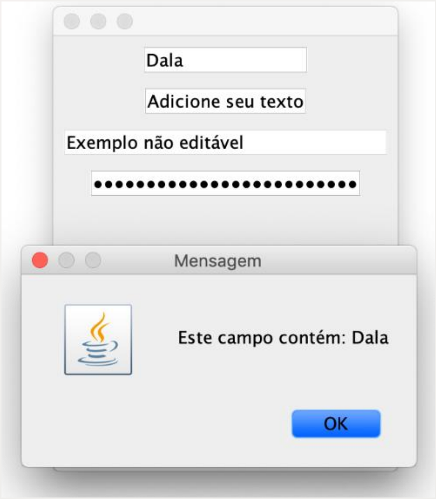
EXEMPLO 3

```
import java.awt.FlowLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.JFrame;
import javax.swing.JOptionPane;
import javax.swing.JPasswordField;
import javax.swing.JTextField;

public class Exemplo3 extends JFrame {

  private final JTextField tf1;
  private final JTextField tf2;
  private final JTextField tf3;
  private final JPasswordField pf1;

  public Exemplo3(){
     setLayout(new FlowLayout());

     tf1 = new JTextField(10);
     add(tf1);
     tf2 = new JTextField("Adicione seu texto");
     add(tf2);
     tf3 = new JTextField("Exemplo não editável", 20);
     tf3.setEditable(false);
     add(tf3);
     pf1 = new JPasswordField("Texto em formato de senha");
     add(pf1);
```

```
TratamentoEventos tEventos = new TratamentoEventos();
tf1.addActionListener(tEventos);
tf2.addActionListener(tEventos);
tf3.addActionListener(tEventos);
pf1.addActionListener(tEventos);
}

private class TratamentoEventos implements ActionListener{

@Override
public void actionPerformed(ActionEvent e) {

String string = "";
if(e.getSource() == tf1){
string = String.format("Este campo contém: %s", e.getActionCommand());
} else if (e.getSource() == tf2){
string = String.format("Este campo contém: %s", e.getActionCommand());
} else if (e.getSource() == tf3){
string = String.format("Este campo contém: %s", e.getActionCommand());
} else if (e.getSource() == pf1){
string = String.format("Este campo contém: %s", e.getActionCommand());
}
JOptionPane.showMessageDialog(null, string);
}
}
}
```

```
import javax.swing.JFrame;

public class Exemplo3_teste {

public static void main(String[] args) {
Exemplo3 ex3 = new Exemplo3();
ex3.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
ex3.setSize(260,350);
ex3.setVisible(true);
}
}
```

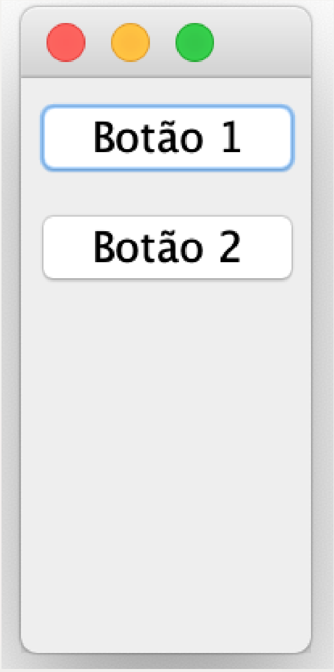


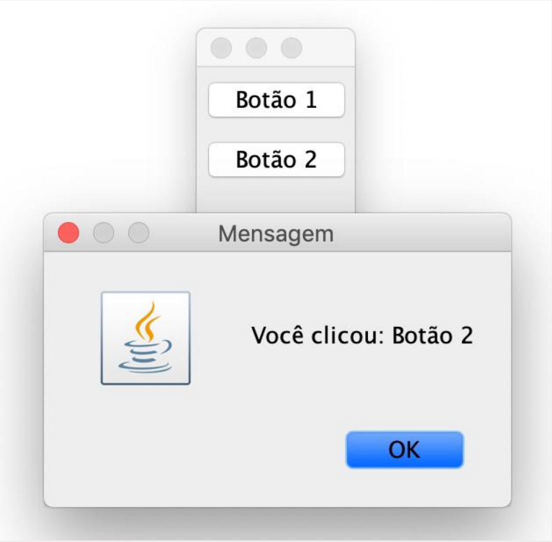


EXEMPLO 4

```
import java.awt.FlowLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JOptionPane;

public class Exemplo4 extends JFrame {

			private final JButton meuBotao1;
			private final JButton meuBotao2;

			public Exemplo4(){
						setLayout(new FlowLayout());
						meuBotao1 = new JButton("Botão 1");
						add(meuBotao1);
						meuBotao2 = new JButton("Botão 2");
						add(meuBotao2);
						TratamentoBotao tb = new TratamentoBotao();
						meuBotao1.addActionListener(tb);
						meuBotao2.addActionListener(tb);
			}

			private class TratamentoBotao implements ActionListener{
						@Override
						public void actionPerformed(ActionEvent e) {

			JOptionPane.showMessageDialog(Exemplo4.this, String.format("Você clicou: %s",
e.getActionCommand()));
						}

			}

}
```

```
import javax.swing.JFrame;

public class Exemplo4_teste {

			public static void main(String[] args) {

						Exemplo4 ex4 = new Exemplo4();
						ex4.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
						ex4.setSize(90,200);
						ex4.setVisible(true);
			}

}
```

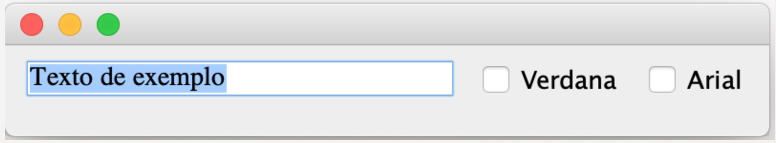




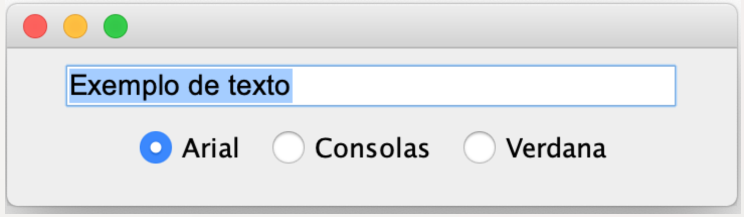
EXEMPLO 5

```
import javax.swing.JFrame;

public class Exemplo5_teste {

        public static void main(String[] args) {
                Exemplo5 ex5 = new Exemplo5();
                ex5.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
                ex5.setSize(400,70);
                ex5.setVisible(true);
        }

}
```

```
import java.awt.FlowLayout;
import java.awt.Font;
import java.awt.event.ItemEvent;
import java.awt.event.ItemListener;
import javax.swing.JCheckBox;
import javax.swing.JFrame;
import javax.swing.JOptionPane;
import javax.swing.JTextField;

public class Exemplo5 extends JFrame {

        private final JTextField tf;
        private final JCheckBox checkbox1;
        private final JCheckBox checkbox2;

        public Exemplo5(){

                setLayout(new FlowLayout());

                tf = new JTextField("Texto de exemplo", 20);
                tf.setFont(new Font("Serif", Font.PLAIN, 14));
                add(tf);

                checkbox1 = new JCheckBox("Verdana");
                checkbox2 = new JCheckBox("Arial");
                add(checkbox1);
                add(checkbox2);

                TratamentoCheckbox tcheck = new TratamentoCheckbox();
                checkbox1.addItemListener(tcheck);
                checkbox2.addItemListener(tcheck);

        }

        private class TratamentoCheckbox implements ItemListener{

                @Override
                public void itemStateChanged(ItemEvent e) {
                        Font font = null;
                        if (checkbox1.isSelected() && checkbox2.isSelected()){
                JOptionPane.showMessageDialog(null, "Opção não disponível ao selecionar os dois itens. Desmarque uma das opções.");
                        } else if (checkbox1.isSelected()){
                                font = new Font("Verdana", Font.PLAIN, 14);
                        } else if (checkbox2.isSelected()){
                                font = new Font("Arial", Font.PLAIN, 14);
                        }

                        tf.setFont(font);
                }
        }
}
```



EXEMPLO 6 - JRADIOBUTTON



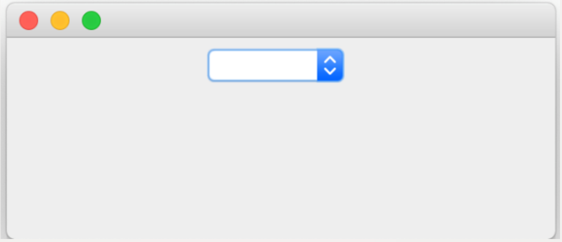

```
import java.awt.FlowLayout;
import java.awt.Font;
import java.awt.event.ItemListener;
import java.awt.event.ItemEvent;
import javax.swing.JFrame;
import javax.swing.JTextField;
import javax.swing.JRadioButton;
import javax.swing.ButtonGroup;

public class Exemplo6 extends JFrame
{
  private JTextField textField;
  private Font arialFont;
  private Font consolasFont;
  private Font verdanaFont;
  private JRadioButton arialRB;
  private JRadioButton consolasRB;
  private JRadioButton verdanaRB;
  private ButtonGroup radioGroup;


  public Exemplo6()
  {
    setLayout(new FlowLayout());

    textField = new JTextField("Exemplo de texto", 25);
    add(textField);

    arialRB = new JRadioButton("Arial", true);
    consolasRB = new JRadioButton("Consolas", false);
    verdanaRB = new JRadioButton("Verdana", false);
    add(arialRB);
    add(consolasRB);
    add(verdanaRB);

    radioGroup = new ButtonGroup();

    radioGroup.add(arialRB);

    radioGroup.add(consolasRB);

    radioGroup.add(verdanaRB);

    arialFont = new Font("Arial", Font.PLAIN, 14);

    consolasFont = new Font("Consolas", Font.PLAIN, 14);

    verdanaFont = new Font("Verdana", Font.PLAIN, 14);

    textField.setFont(arialFont);


    arialRB.addItemListener(new TratamentoRadioButton(arialFont));

    consolasRB.addItemListener(new TratamentoRadioButton(consolasFont));

    verdanaRB.addItemListener(new TratamentoRadioButton(verdanaFont));

  }


  private class TratamentoRadioButton implements ItemListener

  {

    private Font font;


    public TratamentoRadioButton(Font f)

    {

      font = f;

    }


    public void itemStateChanged(ItemEvent event)

    {

      textField.setFont(font);

    }

  }

}
```

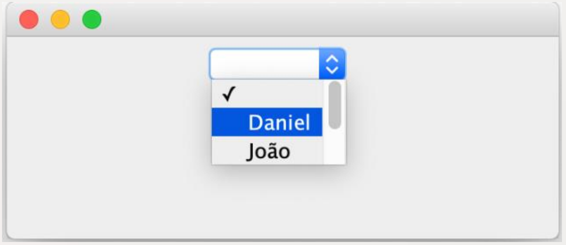
```
import javax.swing.JFrame;

public class Exemplo6_teste
{
  public static void main(String[] args)
  {
    Exemplo6 ex6 = new Exemplo6();
    ex6.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    ex6.setSize(300, 100);
    ex6.setVisible(true);
  }
}
```

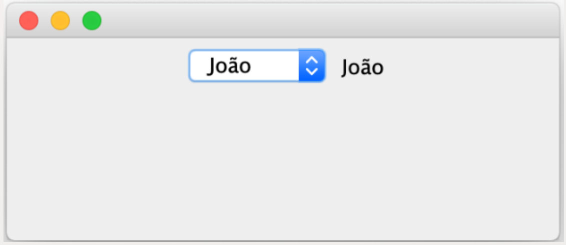
EXEMPLO 7 - JCOMBOBOX

```
import javax.swing.JFrame;

public class Exemplo7_teste
{
  public static void main(String[] args)
  {
                Exemplo7 ex7 = new Exemplo7();

ex7.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
                ex7.setSize(350, 150);
                ex7.setVisible(true);
  }
}
```

```
import java.awt.FlowLayout;
import java.awt.event.ItemListener;
import java.awt.event.ItemEvent;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JComboBox;

public class Exemplo7 extends JFrame
{
  private final JComboBox<String> nomesJComboBox;
  private final JLabel label;

  private static final String[] nomes = {"","Daniel", "João",  "Maria", "Carlos"};

  public Exemplo7(){
    setLayout(new FlowLayout());

    nomesJComboBox = new JComboBox<String>(nomes);
    nomesJComboBox.setMaximumRowCount(3);

    nomesJComboBox.addItemListener(new ItemListener(){
        @Override
        public void itemStateChanged(ItemEvent event)
        {
            if (event.getStateChange() == ItemEvent.SELECTED)
                    label.setText(nomes[nomesJComboBox.getSelectedIndex()]);
        }
    });

    add(nomesJComboBox);
    label = new JLabel();
    add(label);
  }
}
```



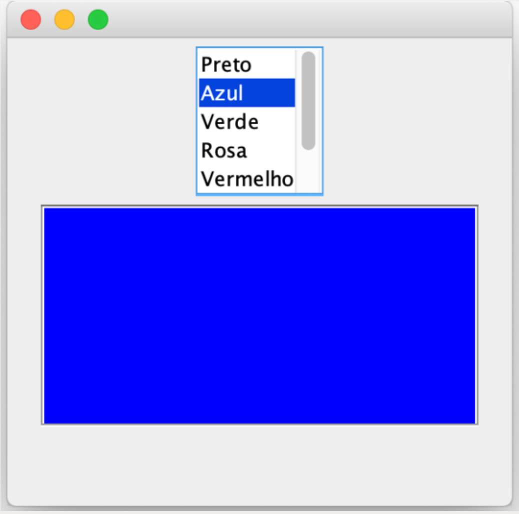




EXEMPLO 8 - JLIST

```
import javax.swing.JFrame;

public class Exemplo8_teste
{
  public static void main(String[] args)
  {
    Exemplo8 ex8 = new Exemplo8();
    ex8.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    ex8.setSize(300, 300);
    ex8.setVisible(true);
  }
}
```

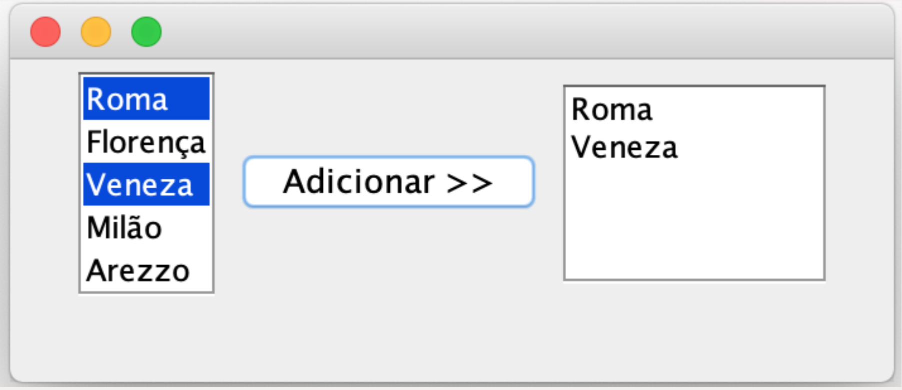
```
import java.awt.FlowLayout;
import java.awt.Color;
import javax.swing.JFrame;
import javax.swing.JList;
import javax.swing.JScrollPane;
import javax.swing.event.ListSelectionListener;
import javax.swing.event.ListSelectionEvent;
import javax.swing.ListSelectionModel;

public class Exemplo8 extends JFrame
{
  private final JList<String> coresJList; // list to display colors
  private final JList<String> pintarJList;
  private static final String[] listaCores = {"Preto", "Azul", "Verde", "Rosa", "Vermelho", "Branco", "Amarelo"};
  private static final Color[] cores = {Color.BLACK, Color.BLUE, Color.GREEN, Color.PINK, Color.RED, Color.WHITE, Color.YELLOW};

  public Exemplo8()
  {
    setLayout(new FlowLayout());

    coresJList = new JList<String>(listaCores);
    pintarJList = new JList<>();

    coresJList.setVisibleRowCount(5);
    coresJList.setSelectionMode(ListSelectionModel.SINGLE_SELECTION);

    add(new JScrollPane(coresJList));
    add(new JScrollPane(pintarJList));

    coresJList.addListSelectionListener(new ListSelectionListener(){
        @Override
        public void valueChanged(ListSelectionEvent event) {
                    pintarJList.setBackground(cores[coresJList.getSelectedIndex()]);
        }
      }
    );
  }
}
```

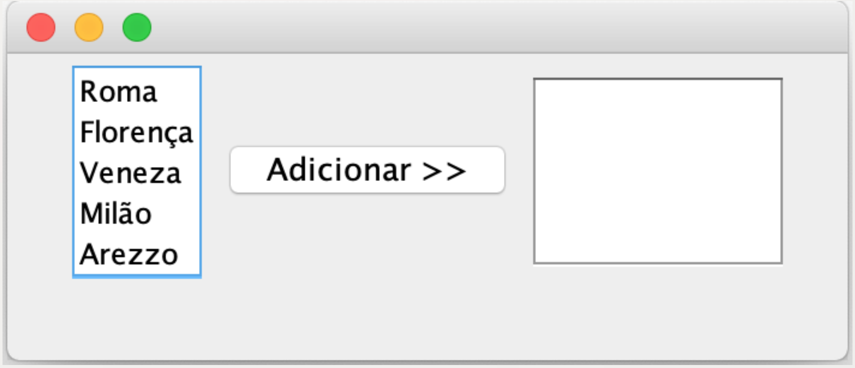


EXEMPLO 9 – JLIST (MULT. SELEÇÃO)

```
import javax.swing.JFrame;

public class Exemplo9_teste
{
  public static void main(String[] args)
  {
    Exemplo9 ex9 = new Exemplo9();

ex9.setDefaultCloseOperation(JFrame.EXIT_ON_CLOS
E);
    ex9.setSize(350, 150);
    ex9.setVisible(true);
  }
}
```

```
import java.awt.FlowLayout;
import java.awt.event.ActionListener;
import java.awt.event.ActionEvent;
import javax.swing.JFrame;
import javax.swing.JList;
import javax.swing.JButton;
import javax.swing.JScrollPane;
import javax.swing.ListSelectionModel;

public class Exemplo9 extends JFrame
{
  private final JList<String> cidades;
  private final JList<String> cidadesConfirmadas;
  private JButton btnAdicionar;
  private static final String[] nomeCidades = {"Roma", "Florença", "Veneza","Milão", "Arezzo"};

  public Exemplo9()
  {
    setLayout(new FlowLayout());

    cidades = new JList<String>(nomeCidades);
    cidades.setVisibleRowCount(5);
    cidades.setSelectionMode(ListSelectionModel.MULTIPLE_INTERVAL_SELECTION);
    add(new JScrollPane(cidades));

    btnAdicionar = new JButton("Adicionar >>");
    btnAdicionar.addActionListener(new ActionListener(){
        @Override
        public void actionPerformed(ActionEvent event){
                    cidadesConfirmadas.setListData(cidades.getSelectedValuesList().toArray(new String[0]));
        }
      }
    );

    add(btnAdicionar);
    cidadesConfirmadas = new JList<String>();
    cidadesConfirmadas.setVisibleRowCount(5);
    cidadesConfirmadas.setFixedCellWidth(100);
    cidadesConfirmadas.setFixedCellHeight(15);
    cidadesConfirmadas.setSelectionMode(ListSelectionModel.SINGLE_INTERVAL_SELECTION);
    add(new JScrollPane(cidadesConfirmadas));
  }
}
```





EXEMPLO 10 – KEYEVENTS

```
import javax.swing.JFrame;

public class Exemplo10_teste
{
  public static void main(String[] args)
  {
    Exemplo10 ex10 = new Exemplo10();

ex10.setDefaultCloseOperation(JFrame.EXIT_ON_CLOS
E);
    ex10.setSize(350, 100);
    ex10.setVisible(true);
  }
}
```

```
import java.awt.Color;
import java.awt.event.KeyListener;
import java.awt.event.KeyEvent;
import javax.swing.JFrame;
import javax.swing.JTextArea;

public class Exemplo10 extends JFrame implements KeyListener
{

  private JTextArea textArea;

  public Exemplo10()
  {

    textArea = new JTextArea(10, 15);
    textArea.setText("Aperte uma tecla...");
    textArea.setEnabled(false);
    textArea.setDisabledTextColor(Color.BLACK);
    add(textArea);
    addKeyListener(this);
  }

  @Override
  public void keyPressed(KeyEvent event) {
                        textArea.setText(String.format("Tecla pressionada: %s", KeyEvent.getKeyText(event.getKeyCode())));
  }

  @Override
  public void keyReleased(KeyEvent event) {
                        textArea.setText(String.format("Você pressionou e soltou a tecla: 
%s",KeyEvent.getKeyText(event.getKeyCode())));
  }

  @Override
  public void keyTyped(KeyEvent event){
                        textArea.setText(String.format("Você pressionou a tecla: %s", event.getKeyChar()));
  }

}
```